Albert Einstein: E = mc²

Albert was born in Ulm in south Germany in 1879 and spent his youth in Munich, where his family owned a small shop that manufactured electric machinery. He did not talk until the age of three. As a youth he was curious about nature and able to understand difficult mathematical concepts. At the age of 12 he taught himself geometry.

However, Albert didn't enjoy school. After a break from school when he was 15, he sat for the entrance exam to a Swiss university. Sadly, he failed the Arts section and so he had to finish secondary school in Aarau, Switzerland. Finally, in 1896, he entered the Swiss National Polytechnic University in Zurich in order to become a science teacher. After Einstein graduated in 1900, he became a tutor. Einstein had also renounced his German citizenship in order to avoid military conscription and he adopted Swiss nationality. In 1902 Einstein could not get a position as a lecturer or teacher, so he took a job at the Swiss Patent Office. Although Einstein and Mileva Maric, another student, had a baby (who was adopted) in 1901, they only married in 1903.

In 1905, Einstein published five papers. The first (Generation and transformation of light) suggested that light consisted of quanta, bundles of energy which behaved sometimes like particles (e.g. in reflection) and other times like waves (e.g. in refraction). This contribution to quantum mechanics eventually earned Einstein a Nobel Prize in 1921.

The second paper (Motion of suspended particles and the kinetic theory) provided a mathematical explanation that molecules in liquids moved and a proof that atoms really did exist.

The most famous paper (on the electrodynamics of moving bodies) introduced the Special Theory of Relativity. This was the basis for a new physics: time, space and mass were all connected with each other, no longer separate and fixed, but different, relative to the observer. In a supplement of 1907 Einstein included the famous equation $E = mc^2$, which showed that matter and energy were interchangeable and no longer separate and constant. This formed the basis for nuclear reactors and bombs.

Because Einstein had no association with a university and because his ideas were difficult to understand, he was accepted only slowly by other scientists. Eventually, in 1909, his first academic appointment was at the University of Zurich. In 1911 he moved with his family to Prague, where he became a professor at the German University there. In 1912 he returned to the Swiss National Polytechnic in Zurich. And in 1913, he was appointed director of the Kaiser Wilhelm Institute for Physics in Berlin. Soon after, in 1914, Einstein and Maric divorced.

In the General Theory of Relativity (1917), Einstein explained the variations in the orbital motion of the planets and predicted the bending of light near a star. When the gravitational bending of light was observed during a solar eclipse in 1919, Einstein's fame spread worldwide. In this year Einstein also married Elsa Lowenthal, who had nursed him back to health two years before, when he had collapsed from exhaustion.

When Hitler came to power, Einstein decided to leave Germany for the United States in 1933. He took a position at the Institute for Advanced Study at Princeton, New Jersey.

In 1939 Einstein and several other physicists wrote a letter to President Roosevelt, pointing out the possibility of making an atomic bomb and the likelihood that the German government was working on it already. This letter helped to start the development of the atomic bomb, but Einstein himself played no role in the work and knew nothing about it at the time.

After World War II, Einstein was active in the cause of international disarmament and world government. He continued his active support of Zionism but declined the offer made by leaders of the state of Israel to become president of that country. Einstein died in Princeton in 1955.

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Albert Einstein: $E = mc^2$

19b

Complete these tasks in your science book.

1 Make a timeline starting at 1870 and finishing in 1960. Mark and number every other decade above the line. Mark and label Einstein's age in increments of ten years below the line. Then indicate on the timeline major events in Albert Einstein's life.

Reading on the lines ...

2 In which country was Einstein born?

3 What did Einstein teach himself during his early high school years?

4 In which country did Einstein finish his schooling?

5 Why couldn't Einstein enter university?

6 What job was Einstein interested in after university?

7 Einstein again couldn't secure a desired position. What job did he take up instead?

8 Why did Einstein renounce his German citizenship?

9 What is a quantum (plural: quanta)?

10 Why didn't other scientists accept Einstein's theory straight after publication?

11 What was Einstein's letter to Roosevelt about?

Reading between the lines ...

12 Why would Einstein have taught himself geometry?

13 Why did Einstein study in Switzerland and not Germany?

14 How many years after his paper did Einstein receive a Nobel Prize?

15 Newton defined space and time. In what way did his concepts differ from those proposed by Einstein?

16 According to Einstein, what has to happen to space if time changes?

17 How many years after publishing his famous papers did Einstein begin working in a university?

18 Why was Einstein suddenly famous during 1919?

19 Why did Einstein leave Germany when Hitler came to power, even though he was a Swiss citizen?

20 Einstein was a pacifist. How did he show this in his life?

21 How old was Einstein when he died?

Reading beyond the lines ...

22 Why may Einstein have disliked school?

23 When Einstein was 15 years old, his family lived in Milan, Italy. Why didn't he attend school there?

24 How does light behave like a particle during reflection?

25 What is refraction?

26 How do molecules in liquids move?

27 What is an atom and what is it made of?

28 Who had defined space time and mass before Einstein? How much earlier?

29 What do the letters in the expression $E = mc^2$ represent?

30 The development of the atomic bomb in the USA was known as the Manhattan Project. What role did Einstein play in it?

31 Why would Einstein have been unhappy about his contribution to the development of nuclear energy?

32 Why would Einstein have been invited to become the president of Israel?

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